K818 — Sharpening the sign, and heading off a trap. The surviving chirality/structure does NOT come from the SO(2)-charge of the spinor (that's scalar — Lyra's retraction — and re-attributing chirality to it would be the retracted error in a new guise). It comes from the DOLBEAULT DEGREE ⊗ the g=7 ω-LOCK: the holomorphic (degree-0) Dolbeault sector = one spacetime chirality; the computed $\omega = \gamma^5 \cdot \chi$_int correlates it to a definite INTERNAL chirality ((2,1) doublet vs (1,2) singlet). So the sign is EXTRACTABLE from what's already computed (Elie's ω-factorization) + which degree the physical weight concentrates in (Hotta–Parthasarathy) — more tractable than a fresh Riemann–Roch.

Keeper | 2026-07-22 | Continuing while the team computes the sign. Clarifying WHERE the sign lives (to avoid re-confusing it with the retracted SO(2)-orientation) and making it more tractable.

★ The trap to avoid — the sign is NOT the SO(2)-charge of the spinor

Lyra correctly retracted "the SO(2) orients the chirality": the SO(2) K-factor commutes with SO(5) → scalar on the spinor → (2,1) and (1,2) have the SAME SO(2)-charge. So the SO(2)-charge of the spinor is chirality-blind — it CANNOT be the sign. Re-attributing the surviving chirality to the SO(2)-charge would be the same retracted error in new clothes. The SO(2)'s only role is energy-positivity — selecting the positive-energy (holomorphic discrete series) MODULE. It does not pick the chirality within it.

Where the sign actually comes from — Dolbeault degree ⊗ ω-lock

Two things, both already partly in hand: 1. Born=Bergman → the physical states are the DOLBEAULT-degree-0 (holomorphic) sector (Hotta–Parthasarathy: L^2 cohomology concentrates in one degree; for the holomorphic-discrete-series weight, degree 0 = Hardy). Degree parity = spacetime chirality (γ^5 = (−1)^q). So degree-0 = one definite spacetime chirality. 2. The g=7 ω-lock (computed, Elie/Lyra): $\omega = \gamma^1 \cdots \gamma^7 = \gamma^5 \cdot \chi$_internal = ±1. This LOCKS the spacetime chirality (γ^5) to a definite INTERNAL chirality (χ_int = the (2,1)/(1,2) split from SO(5)/n_C). - Combine: degree-0 → fixes γ^5 → the ω-lock fixes χ_int → **fixes which internal half survives — (2,1) SU(2)_L DOUBLET or (1,2) SINGLET.** THAT is the sign, and it determines whether the surviving fermions are left-doublets (matches SM) or singlets (doesn't).

★ Why this makes the computation MORE tractable

The sign is NOT a fresh, foreign Riemann–Roch. It's: - (a) the ω-factorization $\gamma^5 \cdot \chi$_int — ALREADY COMPUTED (Elie 4776/4777: the relative sign between γ^5 and χ_int in ω), times - (b) which degree the physical SM-fermion weight lands in (Hotta–Parthasarathy) — a weight lookup on the holomorphic discrete series. Chain: physical weight → Dolbeault degree (a→ spacetime chirality) → ω-lock (→ internal chirality) → doublet-or-singlet. So the decisive check reduces to: *does the degree-0 spacetime chirality, ω-locked, land on the (2,1) doublet?* Extractable from the computed ω + the weight.

The honest residual (still candidate)

- The weight of the SM fermion bundle (which homogeneous bundle carries the doublets) still must be pinned → which degree → which spacetime chirality. That's the one input to the chain not yet fixed.
- Whether the ω-lock sends degree-0 to (2,1) or (1,2) is the SIGN — a definite computation from the pieces, not a 50-50, but not yet done.
- CANDIDATE, compute-don't-assert. This clarifies the STRUCTURE of the sign computation (and heads off the SO(2)-charge trap); it does NOT assert the answer. Do NOT re-bank until the chain gives (2,1)-doublet.

For the team (feeds 07-22h)

- The sign = Dolbeault-degree ⊗ ω-lock, NOT the SO(2)-charge. Compute it as: physical weight → degree → γ^5 → (via the computed ω=γ^5·χ_int) → χ_int → (2,1) or (1,2).
- Elie's ω-factorization already gives the γ^5↔χ_int relative sign; combine with Lyra's weight→degree. Don't recompute a fresh index; combine the two pieces.
- Guard: if anyone attributes the chirality to the SO(2)-charge, that's the retracted error — flag it.

— Keeper K818, 2026-07-22. The sign is NOT the spinor's SO(2)-charge (scalar, chirality-blind — retracted-error trap). It's the DOLBEAULT DEGREE ⊗ the g=7 ω-LOCK: degree-0 (holomorphic, Born=Bergman) = one spacetime chirality (γ^5=(-1)^q); ω=γ^5·χ_int (computed) locks it to a definite internal chirality ((2,1) doublet vs (1,2) singlet). Sign = does degree-0, ω-locked, give the (2,1) doublet (=L-doublet, SM) or (1,2) singlet. Extractable from the computed ω-factorization + the Hotta–Parthasarathy weight→degree — more tractable than a fresh Riemann–Roch. CANDIDATE, compute-don't-assert. See [[Keeper_K817_L2_cohomology_concentrates_in_one_degree_Hotta_I 07-22]], Elie 4776/4777 (ω-factorization), Lyra (SO(2)=energy-positivity, retraction).